

Cultural Attributes and Their Influence on Consumption Patterns in Popular Music

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Abstract. In this paper we leverage recent developments in the way scholars access, collect, and analyze data to reexamine consumption dynamics in popular music. Using web-based tools to construct a dataset that distills songs' musical content into a handful of discrete attributes, we test whether and how these attributes affect a song's position on the *Billboard Hot 100* charts. Our analysis suggests that attributes matter, beyond the effect of artist, label, and genre affiliation. We also find evidence that the relational patterns formed between attributes—what we call cultural networks—crowds songs that are too similar to their neighbors, adversely affecting their movement up the charts. These results suggest that culture possesses its own sphere of influence that is partially independent of the actors who produce and consume it.

Keywords: culture, consumption, networks, attributes, music.

1 Introduction

Over the last two decades, a small but growing group of scholars has worked to import and develop new techniques to understand better the antecedents, consequences, and qualities of culture (e.g., DiMaggio 1994; Mohr 1998; Weber 2005; Lizardo 2006; Lena and Peterson 2008). These developments have been bolstered by methodological advancements in “big data” and computer science, which increasingly influence the work of sociologists and organization and management scholars. Despite this bridge between the social and computational sciences, however, issues of empirical measurement—how to operationalize conceptual variables and processes, determine appropriate indicators for them, and specify discrete units to model and test—remain central for scholars interested in the study culture (Mohr and Ghaziani 2014).

In this paper, we engage with these issues by leveraging recent advances in the way we access, collect, and analyze data to reexamine cultural consumption and evaluation in popular music. To do this, we have constructed a unique dataset describing over 25,000 songs that appeared on the *Billboard Hot 100* charts between 1958 and 2013. Published weekly by *Billboard Magazine*, these charts contain information on and rankings of the most popular songs in public circulation. The analyses conducted in this paper test whether chart performance is a function of certain musical characteristics,

measured here using a suite of algorithmically-determined attributes that represent the sonic fingerprint of each track (e.g., loudness, tempo, valence, etc.).¹

Introducing a new means of measuring cultural content in the domain of popular music allows us to investigate two questions that have important implications for the study of cultural production and consumption outcomes. First, we ask to what extent aesthetic attributes affect consumers' evaluations of cultural products. Extant literature on this topic emphasizes the role producers, distributors, critics, and consumers play in determining what kinds of culture is created and how it is received (e.g., Peterson 1990). In other words, factors such as artist familiarity, payola, label size, and peer recommendation are the primary determinants of whether a song is considered a hit or a flop (Dowd 1992; Salganik, Dodds, and Watts 2006; Rossman 2012). While some of the work in this area has hinted that there may be other forces at play, no one has provided compelling evidence to explain how cultural content—the attributes of the cultural products themselves—shapes this process. We propose here that, although people are ultimately responsible for the production and evaluation of new cultural forms and practices, attributes play a significant role in determining if and how cultural products (e.g., songs) are consumed.

Yet even if we agree that cultural attributes play a predictive role in the evaluation process, the way in which such influence asserts itself remains unspecified. To address this puzzle, we ask whether certain patterns of attribute similarity affect the position and performance of cultural products in the competitive marketplace. Specifically, we propose that the relational structures formed between bundles of sonic attributes may affect when and where new songs appear on the charts. In other words, a song's position within its cultural network—the system of relationships defined by shared attributes between songs, and the cultural fabric within which a song is embedded—endogenously influences how that song is evaluated by consumers.

After providing a brief review of relevant work in cultural sociology and network science, we introduce our conception of cultural networks and generate a series of models designed to test whether such relationships exist, and to what effect. Although these analyses are exploratory in nature, they provide preliminary evidence that attributes matter, both independently and in the way they structure songs' relationships to each other. By recognizing the possibility that culture has its own sphere of influence that is partially independent of the actors who produce and consume it, we rethink some of the basic mechanisms associated with cultural production and consumption. Employing endogenous explanations to understand how songs are evaluated suggests that the engine of production and consumption can arise from cultural structures themselves, generating new insights into the dynamics of a multi-billion dollar industry while reinforcing the notion that large-scale quantitative comparison can effectively extend scientific research on culture.

2 Culture, Measurement, and a Move toward Relationality

The heart of culture research, both in the humanities and social sciences, lies in its contextual richness and interpretive complexity. Much of the research in this area has

¹ More information about these attributes, and how they are constructed, can be found in the Data & Methods section of this paper and in the Appendix.